JAN-G-383

22 AUGUST 1946

JOINT ARMY-NAVY SPECIFICATION

GILDING METAL (90/10 BRASS); SHEET AND STRIP

Army Number	Navy Number
57-171-1C	47G2

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army	Navy Department
57-171-1B	
20 Feb. 1941	

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A-1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material[1] together with Appendix II, Metals, Part A, Definitions and Tests.

FEDERAL SPECIFICATION

QQ-M-151—Metals; General Specification for Inspection of.

A-2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[1]

B. GRADE, FORMS AND CONDITIONS

B-1. *Grade and forms.*—Gilding metal shall be furnished in one grade and in the following forms, as specified (see par. H-1):

Sheet
Strip

B-2. *Conditions.*—Gilding metal shall be furnished in the following conditions, as specified (see par. H-1):

Annealed:
- Light anneal
- Intermediate anneal
- Drawing anneal

Cold rolled:
- Quarter hard
- Half hard
- Three-quarter hard
- Hard
- Extra hard
- Spring
- Extra spring

[1] Applicable only to Navy purchases.

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C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—Gilding metal shall be made from high grade copper and high grade zinc or from high grade ingot metals and clean scrap of known composition, which will produce material meeting the requirements of this specification.

C–2. *Workmanship.*—

C–2a. The metal shall be free from oil, grease, oxidation, laminations, double metal, scale, chips, dirt or grit, dented or bent edges, pipes, slivers, laps, cracks, deep scratches, wrinkles, and other defects which will render the metal unsuitable for the purpose intended. The metal shall be clean and free from acid.

C–2b. The metal shall be uniform in thickness, composition, anneal or temper, and free from kinks and warps. Coiled material shall be loose enough to feed satisfactorily in the cupping machine.

C–2c. Sheet and strip, when ordered for spinning or drawing, shall be of such quality as to present, after spinning or drawing, a good smooth surface that will not be spotted or open grained.

C–2d. Sheet and strip, when ordered for either shallow or deep cupping, shall not have directional properties which result in the formation of ears when the material is cupped.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Chemical composition.*—

E–1a. Gilding metal shall meet the requirements shown in table I.

TABLE I.—*Chemical requirements.*

Percent Composition				
Copper	Lead (Maximum)	Iron (Maximum)	Total Elements other than Cu and Zn (Maximum)	Zinc
89.0—91.0	0.05	0.05	0.13	Remainder

E–1b. An analysis of each lot shall be furnished by the contractor, showing the percentage of each element designated in table I.

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E-2. *Physical requirements—annealed gilding metal.*—When gilding metal, 0.020 inch and over in thickness, is ordered annealed, it shall meet the physical requirements shown in table II for the anneal and thickness specified.

TABLE II.—*Physical requirements of annealed sheet and strip 0.020 inch and over in thickness.*

Anneal and Thickness, inches	Tensile strength psi., (Minimum)	Elongation in 2 inches (Minimum) Percent
Light anneal:		
0.020 to 0.050, incl.	38,000	25
over 0.050 to 0.100, incl.	38,000	27
over 0.100	38,000	30
Intermediate anneal:		
0.020 to 0.050, incl.	36,000	30
over 0.050 to 0.100, incl.	36,000	32
over 0.100	36,000	35
Drawing anneal:		
0.020 to 0.050, incl.	34,000	35
over 0.050 to 0.100, incl.	34,000	38
over 0.100	34,000	40

E-3. *Grain size—annealed gilding metal.*—When specified, gilding metal under 0.020 inch in thickness ordered annealed shall meet the grain size requirements shown in table III for the anneal specified. In addition to the tensile requirements shown in table II for annealed gilding metal 0.020 inch and over in thickness, grain size may be specified in which case the grain size of the material shall meet the requirements shown in table III for the anneal specified.

TABLE III.—*Grain size requirements.*

Anneal	Grain size, millimeter		
	Nominal	Minimum	Maximum
Light anneal	0.015	(1)	0.025
Intermediate anneal	.025	0.015	.040
Drawing anneal	.035	.025	.050

(1) Although no minimum grain size is required, the material shall be fully recrystallized.

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E-4. *Tensile strength—cold-rolled gilding metal.*—Cold-rolled gilding metal shall meet the tensile strength requirements shown in table IV.

TABLE IV.—*Tensile strength requirements of cold-rolled gilding metal.*[1]

Condition (Temper)	Tensile strength (psi)	
	Minimum	Maximum
Quarter hard	40,000	50,000
Half hard	47,000	57,000
Three-quarter hard	52,000	62,000
Hard	57,000	66,000
Extra hard	64,000	72,000
Spring	69,000	77,000
Extra spring	72,000	80,000

[1] Corresponding Rockwell hardness numbers are shown in paragraph II-2 for information only.

E-5. *Dimensions and permissible variations.*—

E-5a. The material shall be furnished in flat lengths or coils, as specified (see par. H-1). The permissible variation at any point in thickness of sheet or strip, to be used for general purposes, shall be shown in table V.

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E–5b. For the manufacture of components for military ammunition where departures from table V are necessary, dimensions shall be as specified in the contract or order (see par. H–1).

TABLE V.—*Permissible variations in thickness.*

Thickness, inches	Permissible variations, inch, plus and minus						
	8 inch and under in width	Over 8 to 14 inch inclusive in width	Over 14 to 20 inch inclusive in width	Over 20 to 28 inch inclusive in width	Over 28 to 36 inch inclusive in width	Over 36 to 48 inch inclusive in width	Over 48 to 60 inch inclusive in width
0.004 and under	0.0003	0.0006	....	....	....	....	....
Over 0.004 to 0.006 incl.	0.0004	0.0008	0.0012	....	....	....	....
Over 0.006 to 0.009 incl.	0.0006	0.0010	0.0015	....	....	....	....
Over 0.009 to 0.013 incl.	0.0008	0.0013	0.0018	0.0025	0.003	0.0035	0.004
Over 0.013 to 0.017 incl.	0.0010	0.0015	0.002	0.0025	0.003	0.0035	0.0045
Over 0.017 to 0.021 incl.	0.0013	0.0018	0.002	0.003	0.0035	0.004	0.005
Over 0.021 to 0.026 incl.	0.0015	0.002	0.0025	0.003	0.0035	0.004	0.005
Over 0.026 to 0.037 incl.	0.002	0.002	0.0025	0.0035	0.004	0.005	0.006
Over 0.037 to 0.050 incl.	0.002	0.0025	0.003	0.004	0.005	0.006	0.007
Over 0.050 to 0.073 incl.	0.0025	0.003	0.0035	0.005	0.006	0.007	0.008
Over 0.073 to 0.130 incl.	0.003	0.0035	0.004	0.006	0.007	0.008	0.010
Over 0.130 to 0.187 incl.	0.0035	0.004	0.0045	0.007	0.008	0.0010	0.012

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E–5c. The width of slit sheet and strip shall not vary from the ordered width by more than the amount shown in table VI.

TABLE VI.—*Permissible variations in width.*

Width, inches	Permissible variations, inch, plus and minus	
	0.004 to 0.032 inch inclusive Thickness	Over 0.032 to 0.187 inch inclusive in Thickness
2 inch and under	0.005	0.010
Over 2 to 8, inclusive	0.008	0.013
Over 8 to 14, inclusive	0.010	0.015
Over 14 to 20, inclusive	0.013	0.018

E–5d. The width of square sheared metal shall not vary from the ordered width by more than the amount shown in table VII.

TABLE VII.—*Permissible variations in width.*
(Square sheared metal up to 120 inches inclusive in length).

Width, inches	Permissible variations, inch, plus and minus		
	1/16 inch and under in thickness	Over 1/16 to 1/8 inch incl. in thickness	Over 1/8 inch in thickness
20 and under	1/32	3/64	1/16
Over 20 to 36 incl.	3/64	3/64	1/16
Over 36 to 60 incl.	1/16	1/16	1/16

E–5e. Permissible variations in width of sawed metal in all gages shall be plus 0.032 and minus 0.000 inch.

E–5f. Permissible variations in width of rolled-edge metal shall be as shown in table VIII

TABLE VIII.—*Permissible variations in width (rolled edge metal).*

Width, inches	Permissible variation, inch, plus and minus		
	0.032 inch and under in thickness	Over 0.032 to 0.125 inch incl. in thickness	Over 0.125 inch in thickness
1/8 and under	0.002	0.002	....
Over 1/8 to 3/8, incl	0.003	0.003	0.004
Over 3/8 to 1, incl.	0.004	0.004	0.006
Over 1 to 2, incl.	0.005	0.007	0.010
Over 2 to 4, incl.	....	0.015	0.015
Over 4 to 6, incl.	....	0.020	0.020
Over 6	....	0.025	0.025

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E–6. *Straightness.*—

E–6a. The edgewise curvature (depth of arc) in any 72 inch length of sheet or strip shipped flat shall not exceed 0.5 inch.

E–6b. The edgewise curvature (depth of arc) in any 72 inch length of coil shall not exceed 1 inch.

E–7. *Length.*—Sheet and strip may be ordered to exact lengths in all sizes, in which case a variation of plus 0.250 inch in length will be permitted.

E–7a. When sheet and strip 12 inches or less wide are ordered in stock lengths, the pieces shipped shall be cut to the nominal or stock length plus or minus one-half inch as a maximum. The shortest acceptable lengths, the maximum permissible percentage by weight of short lengths, and the required minimum percentage by weight of stock lengths in any one shipment shall be as shown in table IX.

TABLE IX.—*Permissible variations in length.*

Nominal or stock length ordered (feet)	Required minimum percentage by weight, of stock lengths	Maximum permissible percentage, by weight, of short lengths				
		In lengths 10 feet and over	In 8- to 10-feet lengths	In 6- to 8-feet lengths	In 4- to 6-feet lengths	In 2- to 4-feet lengths
10 to 12	60	40	30	20	10	..
10	60	..	40	30	20	10
8	70	..	..	30	20	10
6	80	..	..	..	20	10

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Inspection.*—Inspection and tests may be made at the place of manufacture or at the point of delivery as specified in the contract or order (see par. H–1).

F–2. *Lot.*—Unless otherwise specified, a lot shall consist of not more than 60,000 pounds of material of the same form, condition and size submitted for inspection at the same time.

F–3. *Selection of samples.*—At least 10 flat lengths or coils shall be selected by the inspector to represent each lot.

F–4. *Gaging.*—Each flat length or coil, selected in accordance with paragraph F–3 shall be gaged for dimensions. Measurements for thickness shall be made over the entire width of the flat length or coil, and the thickness shall be measured in at least three different places, not less than 1 foot from each end and at such other points as may be selected by the inspector.

F–5. *Sample for Government analysis.*—

F–5a. From each flat length or coil, selected in accordance with paragraph F–3, the inspector shall obtain an equal amount of metal which shall be thoroughly mixed, and not less than 2 ounces of the resultant mixture shall be used for chemical analysis.

F–5b. Drillings, millings, or sawings shall be taken on material 0.040 inch or more in thickness.

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F-5c. The drill, cutter, saw, or other tool used for sampling the metal shall be thoroughly cleaned. No lubricant shall be used in the operation, and the drillings, millings, sawings or clippings shall be carefully treated with a magnet to remove any particles of iron introduced in taking the sample.

F-6. *Government analysis.*—

F-6a. At the option of the purchaser, analysis for determining elements other than copper and zinc may be made by the spectrographic method, provided that in case of dispute, analysis by standard gravimetric or volumetric methods shall be the basis for acceptance.

F-6b. Analysis shall regularly be made only for copper, lead, and iron. If, however, the presence of elements other than copper, lead, iron, and zinc is suspected or indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements is not in excess of the limits specified.

F-7. *Tension tests.*—

F-7a. One tension test shall be made for each 5,000 pounds of material or fraction thereof in a lot, provided that not less than a total of three tests shall be made for each lot.

F-7b. The axis of the tension test specimens shall be parallel to the direction of rolling.

F-7c. Wherever practicable, test specimens shall be machined to the form and dimensions of figure 5, Federal Specification QQ-M-151, or Appendix II, Metals, Part A. Where the material is of insufficient width to provide a standard specimen, the specimen shall be slightly necked as required by the Inspector.

F-7d. To be acceptable the results obtained in the tension tests shall comply with the limits specified in table IV, but one test shall be allowed outside the limits with maximum variation of 1,000 pounds per square inch from the limits.

F-8. *Grain size determination.*—

F-8a. The ends of tension test specimens before pulling may be used for grain size determination. At least three grain size determinations shall be made for each lot, using specimens taken from different sample flat lengths or coils. The grain size shall be determined on a plane parallel to the surface of the flat lengths or coils.

F-8b. The grain size shall be determined by direct comparison with standard micrographs or by procedure outlined in paragraph F-8c. In case of dispute, the diameter of the average grain shall be determined as prescribed in paragraph F-8c. In counting individual grains, the original crystal, including twinned layers, shall constitute one grain.

F-8c. *Method of grain-size measurement.*—

F-8c(1). A circle 79.8 mm. in diameter (or a rectangle of the same area, 5,000 square mm.) shall be drawn on the rough side of a ground-glass screen, the center of the circle being near the center of the rectangular section of the screen. This ground glass

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shall be mounted in a frame which fits a metallographic camera, the smooth side of the glass being on the outside. When the image of the specimen for grain-size determination is focused on the screen, the circle will be plainly visible and its circumference shall be well within the image.

F-8c(2). When the image is properly focused the grains intersected by the circumference of the circle (or periphery of the rectangle) shall be checked and counted. Since the check marks must be made on the smooth side of the glass, a soft red pencil, such as used in laboratories for marking beakers and flasks, will be found satisfactory. The marks used to indicate the boundary grains are usually short, straight lines intersecting the circumference of the circle and perpendicular to it. The completely included grains shall be next checked and counted, after which the red marks may be erased from the glass with a dry cloth. The specimen may then be moved and other measurements made as desired.

F-8c(3). One-half the number of grains intersected by circumference of the circle (or periphery of the rectangle), added to the number of completely included grains, gives the number of equivalent whole grains within the circle.

F-8c(4). *Formulas.*—The following formulas shall be used in computing the grain size:

$$X = \tfrac{1}{2}w + z$$

$$f = \frac{m^2}{5{,}000}$$

$$n = fx$$

$$d = \frac{1}{\sqrt{n}}$$

where

w = number of boundary grains,
z = number of completely included grains,
x = equivalent number of whole grains in 5,000 square millimeters,
m = magnification used,
f = multiplier to obtain number of grains per square millimeter,
n = number of grains per square millimeter,
d = diameter of average grain in millimeters.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G-1. *Packing.*—

G-1a. All material shall be properly segregated as to size when loaded for shipment.

G-1b. Flat lengths and coils shall be packed in boxes or crates so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery.

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G–2. *Marking.*—Each shipping container shall be marked with the name of the material, form, condition, size, and the weight contained therein, as defined by the contract or purchase order under which shipment is made, the name of the contractor, and the number of the contract or order. In addition, shipments for the Navy shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. Requests, requisitions, schedules, and contracts or purchase orders should cover the following features:

(a) Title, number, and date of this specification.
(b) Form, condition and dimensions of gilding metal required (see pars. B–1, B–2, and E–5).
(c) Whether cut flat lengths or coils are desired (see par. E–5a).
(d) Special tolerances if required (see par. E–5b).
(e) Place of inspection and tests (see par. F–1).

H–2. The conditions (tempers) of rolled gilding metal with equivalent reductions and Rockwell hardness values corresponding to the tensile requirements in table IV are as follows:

Condition (Temper)	Nominal reduction (B. & S. gage No.)	Approximate reduction (percent)	B scale[1]		Superficial[2] 30-T Scale	
			Min.	Max.	Min.	Max.
Quarter hard	1	10.9	27	56	34	54
Half hard	2	20.7	50	66	50	61
Three-quarter hard	3	29.4	59	71	56	64
Hard	4	37.1	65	75	60	67
Extra hard	6	50.0	72	79	65	70
Spring	8	60.5	76	81	68	71
Extra spring	10	68.7	78	83	69	72

[1] The B scale applies to metal 0.020 inch in thickness and over.
[2] The 30-T scale applies to metal 0.012 inch in thickness and over.

H–3. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this index may be obtained upon application to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–4. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and

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H-5. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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